Physical Architecture — PD-RPM Remote Care System

An architecture is "the structure of components, their relationships, and the principles and guidelines governing their design and evolution over time." The **physical architecture view** expresses that structure as the physical elements of the system and their interfaces. Figure 1 is the PD-RPM physical architecture, organized by the four physical locations the system actually occupies, and corresponds to DoDAF **SV-1** (systems interface description) and **SV-2** (systems resource flow). The element inventory and interface matrix on page 2 complete it as an **SV-6**.

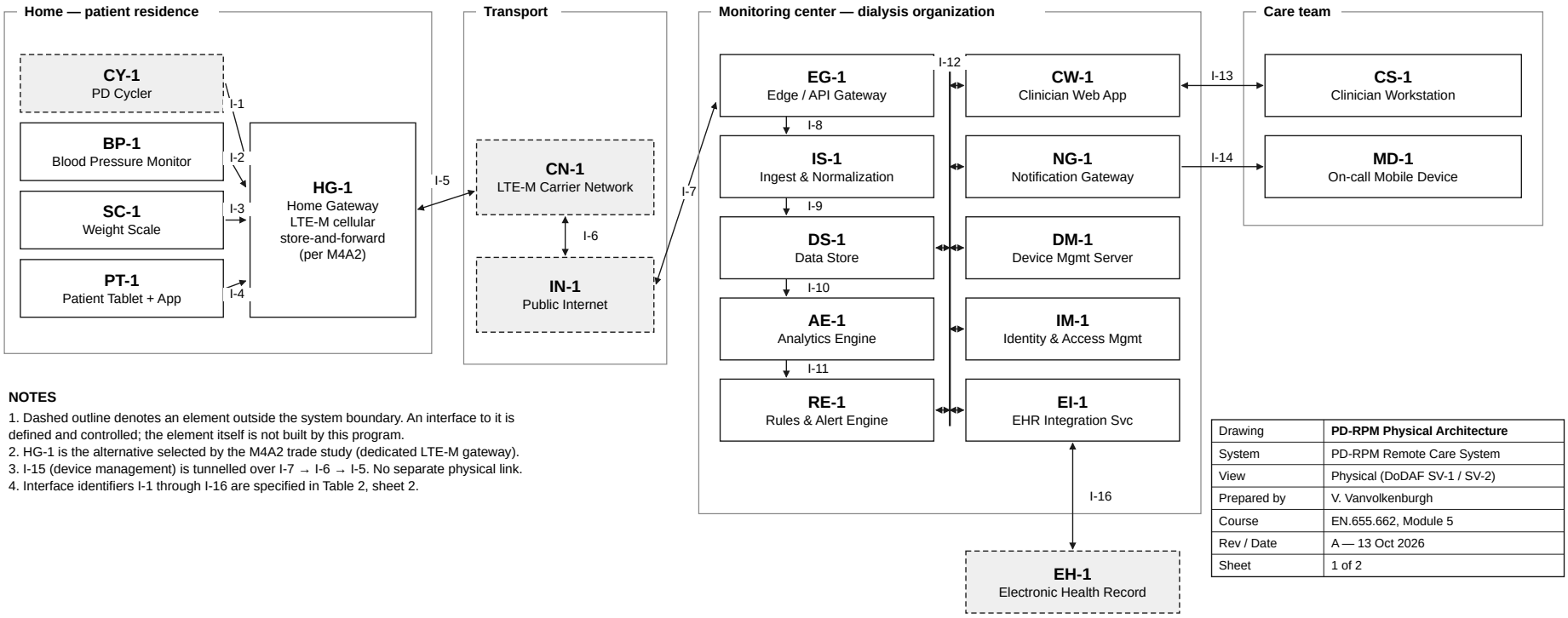


Figure 1. PD-RPM physical architecture (SV-1 / SV-2). Twenty physical elements in four locations; sixteen interfaces, each tagged with the identifier used in Table 2.

Principles governing design and evolution. Four rules constrain every element above. One, nothing inside the home makes a clinical judgment — HG-1 buffers and forwards, and all evaluation happens at RE-1 under version control. Two, every interface crossing a location boundary is mutually authenticated and encrypted, so adding a location adds no new trust assumption. Three, monitoring-center services are stateless behind the I-12 bus, so any one can be replaced without touching its neighbors. Four, CY-1 and EH-1 stay outside the boundary, so their vendors' release cycles cannot stall this program.

On mental models. Drawing the physical view forced an assumption into the open that the functional view had concealed: I had been picturing "the monitoring center" as one thing. It is ten services with five distinct failure modes, and the patient cannot tell which one failed. Making the picture explicit is what let that be examined rather than carried.

Element Inventory and Interface Matrix — PD-RPM Remote Care System

Table 1. Physical element inventory

ID	Physical element	Type	Segment	M4A1 functions allocated
CY-1	PD Cycler	HW (external)	Home	— (source)
BP-1	Blood Pressure Monitor	HW	Home	1
SC-1	Weight Scale	HW	Home	1
PT-1	Patient Tablet + App	HW / SW	Home	1, 3
HG-1	Home Gateway, LTE-M	HW / FW	Home	1, 2, 4, 5, 6
CN-1	LTE-M Carrier Network	Svc (external)	Transport	6
IN-1	Public Internet	Svc (external)	Transport	6
EG-1	Edge / API Gateway	SW	Monitoring	7
IS-1	Ingest & Normalization	SW	Monitoring	7, 8
DS-1	Data Store (time series + relational)	SW	Monitoring	8
AE-1	Analytics Engine	SW	Monitoring	9
RE-1	Rules & Alert Engine	SW	Monitoring	10
CW-1	Clinician Web Application	SW	Monitoring	11
NG-1	Notification Gateway	SW	Monitoring	12
DM-1	Device Management Server	SW	Monitoring	sustains 1–6
IM-1	Identity & Access Management	SW	Monitoring	sustains 11, 12
EI-1	EHR Integration Service	SW	Monitoring	8, 11
CS-1	Clinician Workstation	HW (COTS)	Care team	11
MD-1	On-call Mobile Device	HW (COTS)	Care team	12
EH-1	Electronic Health Record	SW (external)	External	— (sink/source)

Table 2. Interface matrix (SV-6)

ID	Interface	Medium / protocol	Data exchanged
I-1	CY-1 → HG-1	USB 2.0 / RS-232 data port	Treatment session record: fill and drain volumes, dwell times, net UF, alarm codes
I-2	BP-1 → HG-1	Bluetooth LE 5.0, Blood Pressure Profile	Systolic, diastolic, pulse, measurement timestamp
I-3	SC-1 → HG-1	Bluetooth LE 5.0, Weight Scale Profile	Body weight, measurement timestamp
I-4	PT-1 → HG-1	802.11n local Wi-Fi (gateway-hosted AP), HTTPS	Daily symptom and exit-site entries
I-5	HG-1 → CN-1	LTE-M (Cat-M1) radio, private carrier APN	Encrypted session payload, gateway telemetry (bidirectional)
I-6	CN-1 → IN-1	IP transit	Routed traffic (bidirectional)
I-7	IN-1 → EG-1	TLS 1.3, mutual certificate authentication	Session upload, delivery acknowledgment (bidirectional)
I-8	EG-1 → IS-1	Internal REST over mTLS	Raw session payload
I-9	IS-1 → DS-1	Internal API over mTLS	Normalized, unit-consistent records
I-10	DS-1 → AE-1	Internal API over mTLS	Patient time series
I-11	AE-1 → RE-1	Internal API over mTLS	Trend series with derived values
I-12	monitoring-center services	Internal service bus, mTLS service mesh	Service-to-service calls among DS-1, RE-1, CW-1, NG-1, DM-1, IM-1, EI-1
I-13	CW-1 → CS-1	HTTPS (TLS 1.3), browser session	Triage queue, trend views, clinician dispositions (bidirectional)
I-14	NG-1 → MD-1	SMS and push notification over carrier	Page with alert summary and patient identifier
I-15	DM-1 → HG-1	TLS 1.3 tunneled over I-7 → I-6 → I-5	Configuration, firmware images, certificates; device health returned
I-16	EI-1 → EH-1	HL7 v2.5.1 and FHIR R4 over TLS	Demographics and prescription of record in; treatment summaries out